

A Validity Ladder for LLM-Simulated Populations

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Author Note

Data, code and every receipt file are at <https://github.com/pskeough/A-Validity-Ladder-for-LLM-Simulated-Clinical-Populations>.

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Abstract

Behavioral researchers now prompt large language models with demographic personas and use the answers in place of survey respondents, and then check these simulated samples usually receive is a reading of individual answers. A simulated sample supports only the uses whose matching validity checks it has passed. The validity ladder is a protocol of a gate and four rungs. The gate asks how many repeated draws make a persona's score precise enough to read. The rungs ask whether a single draw is an answer pattern a person could give, whether gaps between groups match the population's, whether the level matches, and whether the items hang together as they do in people. Each rung has a statistic, a pass rule, and a use that a pass supports. We demonstrate the ladder on 28,800 PHQ-8 assessments from four models, read against the National Health and Nutrition Examination Survey. Persona means of a few draws are precise enough to read, and every model fails all four rungs above the gate. Single draws are less varied than real respondents at the same total score. The gap between low and high income is steepened in every model. Every model's population mean sits above the survey's after post-stratification. At level 4, two models have no general factor and the other two fail invariance across income. Survey respondents drawn through the same design pass the gate, levels 1 and 3, and the factor checks of level 4, and each planted failure trips the rung it targets. Level 2 rarely fails survey respondents and detects a doubled gap; certifying that a gap is kept needs more draws per persona and a population gap the survey measures precisely. The code, the corpus and a receipt for every reported number are public.

Keywords: silicon samples, large language models, simulated participants, validity, measurement invariance, PHQ-8, NHANES

A Validity Ladder for LLM-Simulated Populations

Large language models (LLMs) prompted with a demographic description will answer a survey as that person might. Researchers have used such answers as *silicon samples* to reproduce vote choice and opinion by group (Argyle et al., 2023; Santurkar et al., 2023), to replicate classic experiments (Park, Schoenegger, & Zhu, 2024), and to pilot instruments before fielding them (Lin, 2026a). For behavioral research, a simulated sample costs little, reaches groups that are hard to recruit, and can be rerun at will.

Each of these uses assumes that the simulated sample stands in for a real one in some respect. How well that assumption is checked varies. An audit of LLM studies across eight social science journals found validation practice “inconsistent and limited” (Desai, Card, & Jacobs, 2026). Park et al. (2024) found GPT-3.5 answering with almost no variation where people vary, and Bisbee, Clinton, Dorff, Kenkel, and Larson (2024) found synthetic opinion distributions narrower than the survey distributions they imitate. Lin (2026b) set out a workflow in which the validity a study needs rises with the claim it makes, and Lin (2026a) list strategies for validating simulated participants against human data. What these sources do not supply is a fixed set of tests, run in a stated order, each with a pass rule and a population reference, that a researcher can apply to one model and report.

One persona from the corpus analysed here shows the problem. Prompted as a single Black woman on Medicaid with a household income under \$35,000, GPT-4o-mini returns a PHQ-8 total of 8 on its first clinical draw. That answer is internally consistent and sits in the mild band. The same prompt run thirty times averages 10.3, above the usual screening cut point. Adults in the national survey who share her race, sex, income band and marital status average 4.7. Reading her answers one at a time would not reveal that the simulated population she belongs to sits well above the real one.

A simulated sample supports only the uses whose matching validity checks it has passed. Checking individual answers covers one of those uses at most, and a researcher who wants group gaps, population levels or item structure from the sample needs a separate

test of each against people who answered the same instrument.

This paper gives the protocol, which we call the validity ladder (Figure 1). A gate measures how many repeated draws a persona needs before its score is precise enough to read. Four rungs then ask whether a single draw is an answer pattern a person could give (level 1), whether gaps between groups match the population’s (level 2), whether the level matches (level 3), and whether the items relate to each other as they do in people (level 4). Each rung is stated as a question, a statistic, a pass rule, and the use that a pass supports. The ladder is read per model, and it records a stop wherever the population reference cannot support a verdict. The ladder makes concrete one of the uses in Lin’s (2026b) workflow, the LLM as a simulated population.

We demonstrate the ladder on 28,800 PHQ-8 assessments from four models, against the National Health and Nutrition Examination Survey (NHANES). We then show that real people pass every rung when run through the same design, and that each rung detects planted failures of known size. Finally, we apply levels 2 and 3 to two silicon-sample datasets released by other authors (Argyle et al., 2023; Meister, Guestrin, & Hashimoto, 2025). The corpus was first reported in an earlier preprint (Keough, 2026). This paper contributes the protocol, the rebuilt analyses at every rung, the controls, and the external demonstration.

The Validity Ladder

The ladder applies to any simulated sample in which a model answers a fixed instrument while conditioned on a persona. We write p for a persona, m for a model, f for a prompt framing and r for one draw, that is, one call to the model and the answer vector it returns. A *cell* is one persona under one model and one framing, and it holds k draws. The *persona mean* is the mean score over a cell’s draws. Figure 1 sets out the gate and the four rungs.

At every rung, “pass” means that the rung’s test detected no failure at the resolution the data allow, and each verdict states that resolution. Every verdict is read per

model, because a researcher runs one model, and a pooled verdict can average models that err in opposite directions. Pooled rows are reported as description. Where the population reference cannot support a verdict, the rung records a stop and names the reason, and it does not report a pass or a fail.

The rungs are ordered by the unit they read and not by nesting. The gate fixes k , the number of draws averaged into a persona mean, and levels 2 and 3 inherit it. A failure at level 1 rules out the uses of a single draw as one respondent, and levels 2 to 4 still run, because they read persona means or the whole simulated population. No other result at one rung stops the rungs above it. The rung order is a reading order, and a pass at level 2 or level 3 is read beside the level-4 invariance result, because a group comparison assumes the items measure the same construct in each group.

The rules were set on the worked example and its reference and run once on a second instrument and reference, BFI-44 answers from three models (Jiang et al., 2024) against the Twin-2K-500 panel (Toubia et al., 2025), before they were frozen. The frozen specification, with every rule, threshold and stop and the code that implements it, is in the public repository (LADDER_SPEC.md, commit 91a1b10 of 2 October 2026), and every analysis of a further dataset applies it unchanged. The worked example therefore illustrates the ladder and does not test it confirmatorily. The thresholds are defaults that a user may change, and each rung gives its source. Supplement S4 lists the multiplicity family and procedure at every rung.

Gate: Regeneration Stability

Question. How many draws must be averaged before a persona’s score is precise enough to read?

Statistic. The gate is a generalizability study (Brennan, 2001; Cronbach, Gleser, Nanda, & Rajaratnam, 1972). Within one model and one framing, the score of draw r of persona p is

$$X_{pr} = \mu + \nu_p + e_{r:p}, \quad (1)$$

Figure 1

The Validity Ladder

Level and question	Unit read	Statistic and pass rule	Comparator	A pass allows
LEVEL 4 Structural fidelity Does the scale hang together the way it does in people, and the same way across groups?	All draws of one model and framing, as one population; personas resampled	One-factor model on polychoric correlations; congruence with the NHANES loadings; invariance across sex, race and income. Pass: in both framings, a general factor (every loading $\geq .30$, first-to-second eigenvalue ratio ≥ 3), congruence $> .95$ at its lower 90% limit and loading RMSD $\leq .10$ at its upper limit, and every invariance step agreeing with NHANES at margin $.08$. The factor within personas is reported, not scored.	NHANES 2005–2018 loadings and invariance verdicts	Treat the total as one score that means the same in every group
LEVEL 3 Population calibration Is each group's level right?	Persona means of k draws, reweighted to each NHANES group's makeup	Simulated minus NHANES mean PHQ-8, for all adults and for nine groups by race, sex and income. Pass: every 90% interval within ± 2 points (0.2 SD and 1 point also reported). The share scoring 10 or more is a second row.	NHANES 2005–2018 group means (2021–2023 beside them)	Use the simulated level as a stand-in for the population's
LEVEL 2 Subgroup fidelity Are the gaps between groups the right size and direction?	Persona means of k draws, differenced within pairs that differ on one attribute	Ratio of the simulated gap to the NHANES gap, with a 98.6% interval (Bonferroni over seven contrasts). Pass: the interval lies within 0.75 to 1.25. Verdicts by the ratio: reversed (below -0.25), missing (-0.25 to 0.25), attenuated (0.25 to 0.75), kept (0.75 to 1.25), steepened (above 1.25). An interval crossing one boundary gets two names, and one crossing more is undetermined. Not read: no population gap; reference too imprecise.	NHANES 2005–2018 gap with the other persona attributes held equal (crude gap beside it)	Read the size and direction of group differences
LEVEL 1 Individual coherence Does a single draw answer like one person?	Single draw: one answer vector, with inference clustered by persona	Person fit of the item pattern under a graded response model fitted to NHANES, compared with adults at the same total. Pass: the shares of draws below the NHANES 5th and above its 95th percentile, each divided by 5%, have 90% intervals within 0.67 to 1.5.	NHANES 2005–2018 adults with the same PHQ-8 total	Use a single draw as one case, such as an exemplar or a test item <i>A failure rules out single-draw uses only. Levels 2 to 4 still run.</i>
GATE Regeneration stability Are persona scores stable enough to read?	Repeated draws of one persona, within one model and framing	Variance of scores between personas and between draws; dependability of a mean of k draws. Pass: k is the fewest draws giving $SE \leq 0.125$ reference SD in each framing (minimum rule 0.25 SD). A single draw passes with $SD \leq 0.25$ reference SD. Dependability is reported as persona separation.	None: draws are compared with each other	Read persona means of k draws at levels 2 and 3


 The gate fixes k , the number of draws averaged into each persona mean, for levels 2 and 3. Apart from this, every level runs whatever the result below it.

Note. Read from the gate at the bottom to level 4 at the top. Each row gives the question a level asks, the unit it reads, its statistic and pass rule, what it is compared with, and the use a pass supports. The gate sets k , the number of draws averaged into each persona mean, for levels 2 and 3. A level-1 failure rules out single-draw uses only, so levels 2 to 4 run whatever level 1 finds. Transgender and multiracial personas have no NHANES group of their own and never enter a group comparison.

with persona variance σ_p^2 and within-cell draw variance σ_r^2 . The dependability of a mean of k draws is

$$\phi(k) = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_r^2/k}, \quad (2)$$

and its standard error is $SE(k) = \sqrt{\sigma_r^2/k}$. The k needed for a target ϕ^* is

$\lceil \phi^*/(1 - \phi^*) \cdot \sigma_r^2/\sigma_p^2 \rceil$, and the k needed for a tolerance t on the standard error is $\lceil \sigma_r^2/t^2 \rceil$.

Draws of one persona carry no order, and a crossed fit of persona by draw index finds no draw main effect, so the absolute and relative coefficients coincide. Components are estimated from expected mean squares and checked against restricted maximum likelihood. Intervals are percentile bootstrap intervals over personas ($B = 1,000$). When a persona's score must also hold across prompt wording, framing becomes a random facet, and $\phi_F(k) = \sigma_p^2/(\sigma_p^2 + \sigma_f^2 + \sigma_{pf}^2 + \sigma_r^2/k)$ has a ceiling that no number of draws can pass.

Pass rule. The gate reads precision. Persona means of k draws may be read when $SE(k)$ is at most 0.25 reference standard deviations in each framing (the minimum rule), and the recommended k brings it to 0.125 standard deviations. A single draw may be read as the persona's score when the within-cell standard deviation is at most 0.25 reference standard deviations. On the PHQ-8 these tolerances are 0.98 and 0.49 points. One point is about the size of the population's sex gap, and half a point keeps a persona mean's error below both that gap and half the low-to-high income gap. When a later level reads the persona mean over both framings, the gate is also reported for that mean. The dependability $\phi(k)$ measures how well persona means separate personas from one another. It is reported beside every verdict, and it is required ($\phi(k) \geq .80$, or .90 for the recommended k) only for uses that compare or rank individual personas. It is not a pass condition for levels 2 and 3, because those levels carry the draw error in their own intervals and because a faithful simulator has the population's small differences between cells: real NHANES respondents drawn through the persona design reach $\phi(30)$ of .63 to .76 across replicates (mean .70). The probability that two draws of one persona fall in different severity categories (band flip) is reported beside the pass rule as a description of

single-draw labels.

What a pass supports. A pass at k supports reading the persona mean of k draws as the model’s answer for that persona, to within the stated error. A pass with $\phi(k) \geq .80$ also supports comparing individual personas with each other. Levels 2 and 3 compare these persona means with the population. The gate says nothing about whether that answer is correct.

Level 1: Individual Coherence

Question. Is a single draw an answer pattern that a real respondent with the same total score could plausibly give, and are the draws as varied as real respondents?

Statistic. A graded response model (Samejima, 1969) is fitted to the population reference with survey weights, and every respondent and every draw is scored on its item parameters. For an answer vector $\mathbf{x}$ with trait estimate $\hat{\theta}$, the log-likelihood person-fit statistic (Drasgow, Levine, & Williams, 1985) is

$$l_0 = \sum_j \sum_k d_{jk} \log P_{jk}(\hat{\theta}), \quad d_{jk} = 1 \text{ if } x_j = k, \quad (3)$$

where P_{jk} is the model’s probability of category k on item j . We standardise it with the correction for an estimated trait (Snijders, 2001) in its polytomous form (Sinharay, 2016), giving l_z^* , with $\hat{\theta}$ the weighted likelihood estimate (Warm, 1989). Low l_z^* marks a vector less likely than the model expects at its trait level (misfit), and high l_z^* marks a vector more regular than expected (overfit). Because l_z^* depends on the total score, each draw is placed in the weighted reference distribution of l_z^* among respondents with the same total, using a randomised probability integral transform. The reference therefore puts exactly 5% of its own respondents below its within-total 5th percentile and 5% above its 95th, whatever the mix of totals. The *misfit ratio* is a model’s share of draws below the within-total 5th percentile divided by .05, and the *overfit ratio* is its share above the 95th percentile divided by .05. A real population has both ratios equal to 1.

Pass rule. A model passes in a framing when the 90% intervals of both ratios lie inside $[1/\tau, \tau]$. The tolerance τ is the smallest value in $\{1.50, 2.00, 3.00\}$ that contains the

departure of every reference subgroup (by sex, race and ethnicity, income band and survey cycle) when each subgroup is scored against the pooled reference. A subgroup's departure is read on the 90% interval of each ratio: it is the interval end nearest 1 when the interval excludes 1, and there is none when the interval covers 1. Reading the interval and not the point ratio keeps the sampling noise of a small subgroup from setting τ . The floor of 1.5 comes from the positive control in Section : real respondents drawn through the persona design pass at 1.5 and not at 1.25, so a reference too small to resolve its subgroups does not set a tighter tolerance than real people meet. On NHANES the largest departure is 1.27, the lower end of the misfit ratio for non-Hispanic Black respondents, so $\tau = 1.5$, and every subgroup passes at that tolerance. Each tail is read on its interval: above the band, below it, inside it, or crossing a boundary. A deficit in both tails, too few atypical and too few highly regular vectors, is called *compressed*. A pass needs both tails, so the rule is an intersection-union test and needs no multiplicity adjustment (Berger, 1982). Intervals come from a persona-clustered bootstrap of the simulated draws, drawn jointly with a Rao–Wu bootstrap over the reference's primary sampling units (Rao & Wu, 1988; Rao, Wu, & Yue, 1992) ($B = 2,000$). For the PHQ-8, the gateway rule (an elevated total with neither of the two core items at 2 or more) is reported at matched totals as a descriptive row. The row is descriptive because, with the corpus's totals kept, uniformly random answer vectors read as too prototypical on it (observed to expected 0.62), while person fit reads them as excess misfit (ratio 6.05).

What a pass supports. A pass supports using single draws as individual cases whose item pattern is as plausible, and as varied, as a real respondent's at the same severity. Examples are exemplar vignettes, item-level test cases for a screening tool, and training data for a classifier that reads item patterns. A compressed failure rules out uses that depend on the spread of presentations, such as stress-testing a screener with atypical cases.

Level 2: Subgroup Fidelity

Question. Is the gap between two groups in the simulated sample the same size and direction as the gap in the population?

Estimand. A simulated contrast compares persona pairs that differ on one attribute and match on the rest, so it is a conditional gap. For a contrast between groups A and B over strata s of the other attributes,

$$g = \frac{1}{S} \sum_{s=1}^S (\bar{X}_{As} - \bar{X}_{Bs}), \quad (4)$$

where $\bar{X}$ is a persona mean. The *standardised* population gap γ is the reference contrast computed within the same strata and averaged with the same equal weights. This gap answers the question the persona design asks, and it is the primary reference. The *marginal* population gap, the crude difference between the two groups in the survey, is reported beside it. The two can differ in sign when the groups differ in composition.

Statistic. The statistic is the ratio $\rho = g/\gamma$. Its interval is the Fieller set (Fieller, 1954): the values c at which the one-sided tests of $\rho > c$ and $\rho < c$, using

$$t_c = \frac{g - c\gamma}{\sqrt{\text{SE}_g^2 + c^2 \text{SE}_\gamma^2}}, \quad (5)$$

do not reject. The simulation and the survey are independent samples, and degrees of freedom are Welch–Satterthwaite, combining the simulation’s and the survey design’s. The standard error of g is the standard deviation of the persona-pair differences divided by the square root of the number of pairs. This standard error treats differences in the gap between strata as noise, so it is conservative. A draw-only standard error, which holds the persona set fixed, is reported as a sensitivity analysis. The population gap’s standard error is a design-based Taylor linearisation of the whole contrast.

Pass rule. The boundaries -0.25 , 0.25 , 0.75 and 1.25 divide the ratio into five regions, named REVERSED, MISSING, ATTENUATED, KEPT and STEEPENED. The verdict names the region that holds the interval, names two adjacent regions when the interval crosses one boundary, and is UNDETERMINED when it crosses more. A contrast passes when

it is KEPT. A model passes level 2 when every contrast that is not stopped is KEPT, fails when any verdict excludes KEPT, and is otherwise unresolved. Each model is one family of contrasts, and every one-sided test runs at .05 divided by the family size (Bonferroni), so with seven contrasts the intervals are 98.6% intervals. Two stops are checked first. The rung records NO POPULATION GAP when the population gap's own interval covers zero, since a ratio to an unresolved denominator has no meaning. It records REFERENCE TOO IMPRECISE when the population gap's interval is too wide for any simulated gap to be read as KEPT. With k the relative half-width of that interval, $k = t \text{SE}_\gamma / |\gamma|$, even an exactly known simulated gap gives a ratio interval spanning a factor of $(1 + k)/(1 - k)$, and the KEPT region spans a factor of 1.25/0.75, so KEPT is out of reach when $k > 1/4$. A contrast with $k > 1/4$ whose verdict already excludes KEPT keeps that verdict, so an imprecise reference can still show a failure but cannot hold a model at unresolved.

What a pass supports. A pass supports reading the size and direction of the group difference from the simulated sample. A kept gap says nothing about either group's level.

Level 3: Population Calibration

Question. Reweighted to the real composition of a group, does the simulated sample have the group's level?

Statistic. Each persona cell c in a group G receives the weight $p_c = N_c / \sum_{c' \in G} N_{c'}$, where N_c is the weighted count of reference respondents in that cell. The residual is

$$R_G = \sum_{c \in G} p_c (s_c - y_c), \tag{6}$$

where s_c is the persona mean and y_c is the reference mean in the same cell, so the simulated level is post-stratified to the population's composition (Holt & Smith, 1979). The persona cells are fixed by design, so the simulated side's sampling error is the draw sampling within cells, $\text{var}_{\text{sim}}(R_G) = \sum_c p_c^2 \text{var}(s_c)$. The reference side's error, which includes the uncertainty in the weights p_c , comes from a stratified delete-one-PSU jackknife over the

survey design (Rust & Rao, 1996). The two sides are independent, and degrees of freedom follow Satterthwaite with the design degrees of freedom on the reference term (Satterthwaite, 1946). The share of the group scoring at or above the instrument’s cut point is a second row, compared both as a difference in percentage points and as a ratio with a Fieller interval.

Pass rule. A row passes at tolerance δ when two one-sided tests at $\alpha = .05$ reject both $-\delta$ and $+\delta$, that is, when the 90% interval of R_G lies inside $(-\delta, +\delta)$ (Lakens, Scheel, & Isager, 2018; Schuirmann, 1987). A model passes for a use when every group row passes, and this rule is again an intersection-union test. The tolerance is the smallest difference that matters for the stated use, and the researcher names it. For the PHQ-8 we report 0.2 reference standard deviations, 1 point and 2 points, with 2 points (about half a standard deviation) as the named default. The prevalence row uses differences of 2.5 and 5 percentage points and a ratio band of 0.80 to 1.25. The 5-point minimal important change of the PHQ-9 (Löwe, Unützer, Callahan, Perkins, & Kroenke, 2004) is a within-person threshold and is not used as a population tolerance.

What a pass supports. A pass at δ supports using the simulated level as a stand-in for the group’s level, within δ , for this prompt, this persona grid and this survey period. A level-3 pass on the mean does not support prevalence or screening uses, and the prevalence row tests those separately.

Level 4: Structural Fidelity

Question. Do the items hang together as they do in people, and in the same way across groups?

Unit. A model’s simulated population in one framing is every draw across its personas. Its item correlations, like the population’s, mix variation between demographic profiles with variation among respondents who share a profile. The persona is the resampling unit for every interval, because its draws share one prompt.

Statistics and pass rule. Polychoric correlations are estimated by two-step maximum likelihood (Olsson, 1979), and a one-factor model is fitted by unweighted least squares. Level 4 has three scored rules and one diagnostic.

1. *R1, a general factor.* Every loading is at least .30 and the ratio of the first to the second eigenvalue, λ_1/λ_2 , is at least 3. Where the reference itself falls below a threshold, as a weak item of an established scale can, the threshold for that item or ratio is the reference's own value, so a model needs a general factor at least as strong as people's. A population without a general factor fails level 4, and its later rules are reported as description.
2. *R2, loadings like people's.* Tucker's congruence between the model's loadings $\boldsymbol{\lambda}$ and the reference loadings $\boldsymbol{\mu}$,

$$\phi_T = \frac{\sum_i \lambda_i \mu_i}{\sqrt{\sum_i \lambda_i^2 \sum_i \mu_i^2}}, \quad (7)$$

has a lower 90% limit of at least .95 (Lorenzo-Seva & ten Berge, 2006), and the root mean square difference between $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$ has an upper 90% limit of at most .10.

Congruence reads the pattern of the loadings and ignores their size, so a population whose items correlate half as strongly as people's, or more strongly, can still reach a congruence near 1; the second condition reads the size. A difference of .10 in every loading of an eight-item scale with loadings of .70 moves coefficient omega from .885 to .818 (loadings of .60) or .934 (loadings of .80). Both limits come from the persona bootstrap of the model paired with the bootstrap of the reference.

3. *R4, invariance like people's.* Multi-group confirmatory factor models test metric and scalar invariance across sex, race and income. The change in fit between nested steps is read as the root mean square error of approximation of the difference (Savalei, Brace, & Fouladi, 2024),

$$\text{RMSEA}_D = \sqrt{G} \sqrt{\max(0, \Delta F - \nu)/d}, \quad (8)$$

where ΔF is the change in the fit function, d the change in degrees of freedom, G the number of groups and ν the expected ΔF under exact invariance. Draws are clustered in personas and the items are coarse, so ν is estimated by permuting group labels among personas (Jorgensen, Kite, Chen, & Short, 2018). A step holds when the upper limit of its 90% interval is below the margin, fails when the lower limit is above it, and is indeterminate otherwise (Marcoulides & Yuan, 2017; Yuan & Chan, 2016). The margin is .08, because the reference itself fails or is indeterminate on three of its six steps at .05; .05 is a sensitivity analysis. R4 passes when every model step matches the reference’s verdict on the same step, fails when any determinate step differs from it, and is unresolved when no step differs but some are indeterminate. A model with an indeterminate step therefore cannot pass R4, and a design needs enough personas per group to resolve every step.

4. *R3, diagnostic: the source of the factor.* Each population’s item covariance is split into a between-cell part and a pooled within-cell part, where a cell is a persona in the simulation and a demographic profile in the reference. The one-factor model is fitted to the within-cell part. This shows whether the factor lives among draws of one persona, as it lives among people who share a profile, and it feeds the same restriction as level 1.

A model passes level 4 when it passes R1, R2 and R4 in every framing.

What a pass supports. Passing R1 and R2 supports scoring the simulated population’s total as one score with human-like item weights and human-like reliability. Passing R4 adds that a group comparison of the total means the same thing in the simulation as in people. A within-persona factor (R3) would support treating draws as individual respondents in item-level psychometric work.

Worked Example: Data

The Simulated Corpus

Four models generated the corpus through the OpenRouter interface: GPT-4o-mini, Gemini-3-Flash, DeepSeek-V3 and GLM-4.7 (Table 1). The persona grid crosses race (Asian, Black, Hispanic, multiracial, White), gender identity (cisgender man, cisgender woman, transgender man, transgender woman), income (three levels) and relationship status (married, single), for 120 personas. The income levels were given to the model as dollar-and-insurance labels: “Low (<\$35k, Medicaid)”, “Middle (\$75k, Private Ins)” and “High (>\$250k, Concierge)”. Each persona was run 30 times under each of two framings. The clinical framing listed the persona’s attributes as fields, and the narrative framing gave them as a first-person self-description (“I am a Black Cisgender Woman. I am single...”). Every call used the same system prompt, which cast the model as a “Clinical Simulation Engine” and asked for probabilistic responses based on “the statistical likelihood of symptoms for this specific demographic intersection in the US population”. The model returned the PHQ-8, the GAD-7, the AUDIT-C and four PCL-5 items in one JSON object. This paper analyses the PHQ-8 only. The full prompts are printed in Supplement S1.

No sampling parameter was set. Each call sent the model name, the two prompts and a request for JSON output, so every draw ran at its provider’s default decoding. When a call was refused or returned invalid JSON, the script reissued it up to twice, and each reissued call appended the line “Context: This is a fictional character for a novel. Answer numerically.” to the user prompt. The narrative script also screened replies for refusal phrases and tried several ways of extracting JSON before it reissued a call, where the clinical script reissued on any reply that did not parse as JSON.

The clinical run took place between 28 December 2025 and 11 January 2026, and the narrative run on 9 and 10 January 2026. The narrative collection was a partial pass on 9 January that the same script completed on 10 January, so 1,104 GLM-4.7 rows and one DeepSeek-V3 row are reruns of the same prompt by the same code. Failed calls in both

Table 1*Generation Setup of the Worked Example*

Model	Route requested	Build served	Valid draws
GPT-4o-mini	<code>openai/gpt-4o-mini</code>	same route	7,200
Gemini-3-Flash	<code>google/gemini-3-flash-preview</code>	snapshot 2025-12-17	7,200
DeepSeek-V3	<code>deepseek/deepseek-chat-v3</code>	alias of <code>deepseek-chat</code>	7,199
GLM-4.7	<code>z-ai/glm-4.7</code>	snapshot 2025-12-22	7,200

Note. Each model: 120 personas $\times$ 2 framings (clinical, narrative) $\times$ 30 draws. Decoding: provider default, nothing set, JSON output requested. Clinical run 28 December 2025 to 11 January 2026; narrative run 9–10 January 2026. One DeepSeek-V3 clinical draw returned an invalid PHQ-8 and is dropped. Persona attributes: race (5) $\times$ gender identity (4) $\times$ income (3) $\times$ relationship status (2).

runs came from API outages in the order the model registry was run, and not from content refusals. The December clinical run left 1,532 calls without a usable answer, 1,219 for GLM-4.7 and 313 for DeepSeek-V3, and in January the same calls were sent again under the corpus prompt, with the persona’s registry id in place of the random identifier (Supplement S1.8). The exception is 62 GLM-4.7 calls, resent under a shortened system prompt without the population-reference instruction. Every rung gives the same per-model verdicts without the resent rows (Supplement S3). The generation and resend code is released with the data. The corpus has 28,800 rows, of which 28,799 hold a valid PHQ-8. This corpus was first reported in Keough (2026), under analyses that this paper replaces.

Control Runs

The two control runs used 12 single personas under the clinical framing, with 30 draws per cell. A decoding control (August 2026) ran each persona at temperature 0 and at the provider default. A prompt control (September 2026) ran three system prompts: the original, the original with the role and the population-reference instruction removed, and a

first-person template. Neither control reproduces the corpus prompt exactly. Both asked for 20 PCL-5 items where the corpus prompt asked for four, and two of the four routes now serve later builds than the dated December snapshots. The controls are compared with each other and not with the corpus. The corpus analyses were not registered in advance, and the ladder’s rules were frozen in a dated public commit before any further dataset was analysed (Section efsec:ladder). The decoding control’s design and primary outcomes were written to a dated file 21 minutes before its first call, and that file is in the release.

Population Reference

The reference is NHANES 2005–2018 (cycles D to J): 36,274 adults aged 18 or over with all eight PHQ-8 items answered and a positive examination weight. Estimates use the examination weights divided by seven and the masked design variables for strata and primary sampling units (105 strata, 214 PSUs, 109 design degrees of freedom). Race comes from the survey’s race and ethnicity recode, with Asian adults identified from 2011 onward. Income is the ratio of family income to the poverty line, banded as below 1.3, 1.3 to 3.5, and above 3.5. Relationship status is married against widowed, divorced, separated or never married, with cohabiting adults left out in the primary analysis and 18- and 19-year-olds, who were not asked from 2007, counted as single. NHANES 2021–2023 is a sensitivity reference at levels 1 and 3, and the 2017–2020 pre-pandemic file at level 3, each analysed with its own design (Akinbami et al., 2022; National Center for Health Statistics, 2018).

Forty-eight of the 120 personas have a reference cell: the cisgender personas of the four races the survey identifies. NHANES records sex and not gender identity, so transgender personas have no group of their own in the reference. The survey’s multiracial category is merged with American Indian, Alaska Native and Pacific Islander respondents and exists only from 2011, so multiracial personas have none either. Both enter levels 1 and 4, which read the whole simulated population, and neither enters a group comparison at levels 2 and 3. The persona labels state dollar incomes and insurance, and the poverty ratio is the primary mapping. A mapping on household income and insurance type is a

sensitivity analysis (Supplement S3).

Results

Gate

No model may be read from a single draw (Table 2). The draw-to-draw standard deviation is between 1.25 and 2.24 points in every model and framing, above the 0.98-point tolerance. Persona means are readable at the corpus’s 30 draws for every model under both rules: the standard error of a 30-draw mean is 0.23 to 0.41 points. The number of draws needed is 3 to 6 under the minimum rule and 9 for DeepSeek-V3, 10 for Gemini-3-Flash, 13 for GPT-4o-mini and 21 for GLM-4.7 under the recommended rule.

The models differ far more in how well their persona means separate personas. The dependability of one draw runs from .21 for GPT-4o-mini under the narrative framing to .90 for Gemini-3-Flash, and at 30 draws GPT-4o-mini’s narrative persona means reach .887, short of the .90 that ranking individual personas would need. Over both framings, 60 draws, GPT-4o-mini reaches .967. The variance between personas ranges from 0.81 for GPT-4o-mini (narrative) to 20.69 for Gemini-3-Flash (narrative), a 26-fold range. Setting the temperature to 0 in the decoding control cut GPT-4o-mini’s draw variance to 12.5% of its default value and did not raise the variance between personas. At temperature 0 a single GPT-4o-mini draw reaches a dependability of only .75. If a persona’s score has to hold across the two prompt framings, DeepSeek-V3 and GPT-4o-mini cannot reach .90 with any number of draws (ceilings .83 and .88).

Two clinical draws of one persona fall in different PHQ-8 severity categories 35.2% of the time [33.6, 36.8]. This rate does not rank the models on stability. A normal distribution with each cell’s own mean and standard deviation, cut at the category boundaries, reproduces the cell flip rates with $R^2 = .88$, so flipping reflects how close a persona sits to a cut point as much as how much its draws vary.

The worked-case persona shows the same pattern. Her 30 clinical draws run from 7 to 16 with a standard deviation of 2.45, and two of them fall in different severity categories

Table 2*Gate: Precision and Separation of a Persona Mean of k Draws, PHQ-8 Total*

Model	Framing	Variance		SE(30)	Separation		k needed		Band flip (%)
		σ_p^2	σ_r^2		$\phi(1)$	$\phi(30)$	Minimum	Recommended	
DeepSeek-V3	Clinical	2.14	2.03	0.26	.51	.97	3	9	27.0
	Narrative	1.61	1.57	0.23	.51	.97	2	7	22.4
Gemini-3-Flash	Clinical	18.74	2.17	0.27	.90	>.99	3	9	31.6
	Narrative	20.69	2.27	0.27	.90	>.99	3	10	28.7
GPT-4o-mini	Clinical	1.89	3.06	0.32	.38	.95	4	13	37.3
	Narrative	0.81	3.09	0.32	.21	.89	4	13	39.9
GLM-4.7	Clinical	15.04	5.04	0.41	.75	.99	6	21	44.8
	Narrative	20.16	3.55	0.34	.85	.99	4	15	37.6

Note. σ_p^2 = variance between personas; σ_r^2 = variance between draws of one persona, in squared PHQ-8 points. SE(30) = standard error of a 30-draw persona mean, in points. $\phi(k)$ = dependability of a mean of k draws (Equation 2), the separation of personas, required only for uses that compare individual personas. Minimum: the smallest k with $SE(k) \leq 0.25$ NHANES standard deviations (0.98 points). Recommended: the smallest k with $SE(k) \leq 0.125$ standard deviations (0.49 points). Band flip = probability that two draws of one persona fall in different PHQ-8 severity categories, a description of single-draw labels. Persona-bootstrap intervals for every entry are in Supplement S4.

60.5% of the time. Her first draw, 8, sits in the mild band, and her persona mean, 10.3, sits in the moderate band.

Level 1: Individual Coherence

No model passes level 1 in either framing (Table 3). At the same total score, model draws land in neither tail of the human person-fit distribution. NHANES puts 5% of adults below its within-total 5th percentile and 5% above its 95th, and the models put 0.0% to 1.0% below and 0.0% to 4.3% above. DeepSeek-V3, Gemini-3-Flash and GPT-4o-mini are compressed in both framings, with too few atypical and too few highly regular answer patterns. GLM-4.7 has the misfit deficit alone, because its interval for the highly regular tail lies inside the tolerance band in the narrative framing and crosses the band's lower edge in the clinical framing. No verdict differs between the two framings, and every model fails at every tolerance from 1.10 to 3.00. At the loosest tolerance, 3.00, Gemini-3-Flash keeps only its misfit deficit and clinical GPT-4o-mini only its overfit deficit.

The item profile shows where the typicality comes from. At matched totals, the models score 0.16 to 0.51 points higher than people on depressed mood. They score 0.22 to 1.11 points lower on fatigue, one of the somatic items people endorse first. The models also repeat themselves across personas. Two draws from different personas at the same total are identical vectors 4.6 to 39 times as often as two NHANES adults at that total. On the gateway row, the models violate the PHQ-8's core-symptom rule at 0.5% to 21% of the human rate at the same totals.

The worked-case persona's 60 draws sit wholly inside the human distribution. None falls below the within-total 5th percentile, none falls above the 95th, and her mean probability integral transform is .40. None of her 32 elevated draws breaks the gateway rule. Each of her answers is a plausible human pattern, and together they are less varied than people at the same severity.

The verdicts hold on the December rows alone and under an unweighted calibration model. Against the smaller NHANES 2021–2023 sample, DeepSeek-V3 and GPT-4o-mini

Table 3

Level 1: Person Fit of Single Draws Against NHANES Adults at the Same Total

Model	Framing	Below NHANES 5th pct.		Above NHANES 95th pct.		Mid-PIT	Verdict
		%	Ratio 90% CI	%	Ratio 90% CI		
DeepSeek-V3	Clinical	0.3	0.01–0.21	0.0	0.00–0.01	.36	Compressed
	Narrative	0.1	0.00–0.11	0.0	0.00–0.01	.37	Compressed
Gemini-3-Flash	Clinical	0.0	0.00–0.00	0.8	0.11–0.39	.66	Compressed
	Narrative	0.0	0.00–0.00	0.9	0.11–0.42	.65	Compressed
GPT-4o-mini	Clinical	1.0	0.09–0.37	0.0	0.00–0.03	.38	Compressed
	Narrative	0.8	0.09–0.30	0.0	0.00–0.03	.38	Compressed
GLM-4.7	Clinical	0.1	0.01–0.04	3.4	0.55–0.92	.65	Misfit deficit
	Narrative	0.1	0.00–0.04	4.3	0.74–1.15	.65	Misfit deficit
NHANES		5.0		5.0		.50	

Note. Each draw is placed in the weighted NHANES 2005–2018 distribution of the person-fit statistic l_z^* among adults with the same PHQ-8 total. % = share of a model’s draws below the within-total 5th percentile (misfit tail) or above the 95th (overfit tail); NHANES places 5% in each by construction. Ratio = share divided by .05, with a 90% interval from a persona-clustered bootstrap drawn jointly with a Rao–Wu bootstrap of NHANES. A model passes when both ratio intervals lie inside 0.67 to 1.5 ($\tau = 1.5$). Mid-PIT = mean probability integral transform, .50 for a population like NHANES. Compressed = a deficit in both tails.

move from compressed to an overfit deficit, because their misfit tails become too imprecise to place (Supplement S3).

Level 2: Subgroup Fidelity

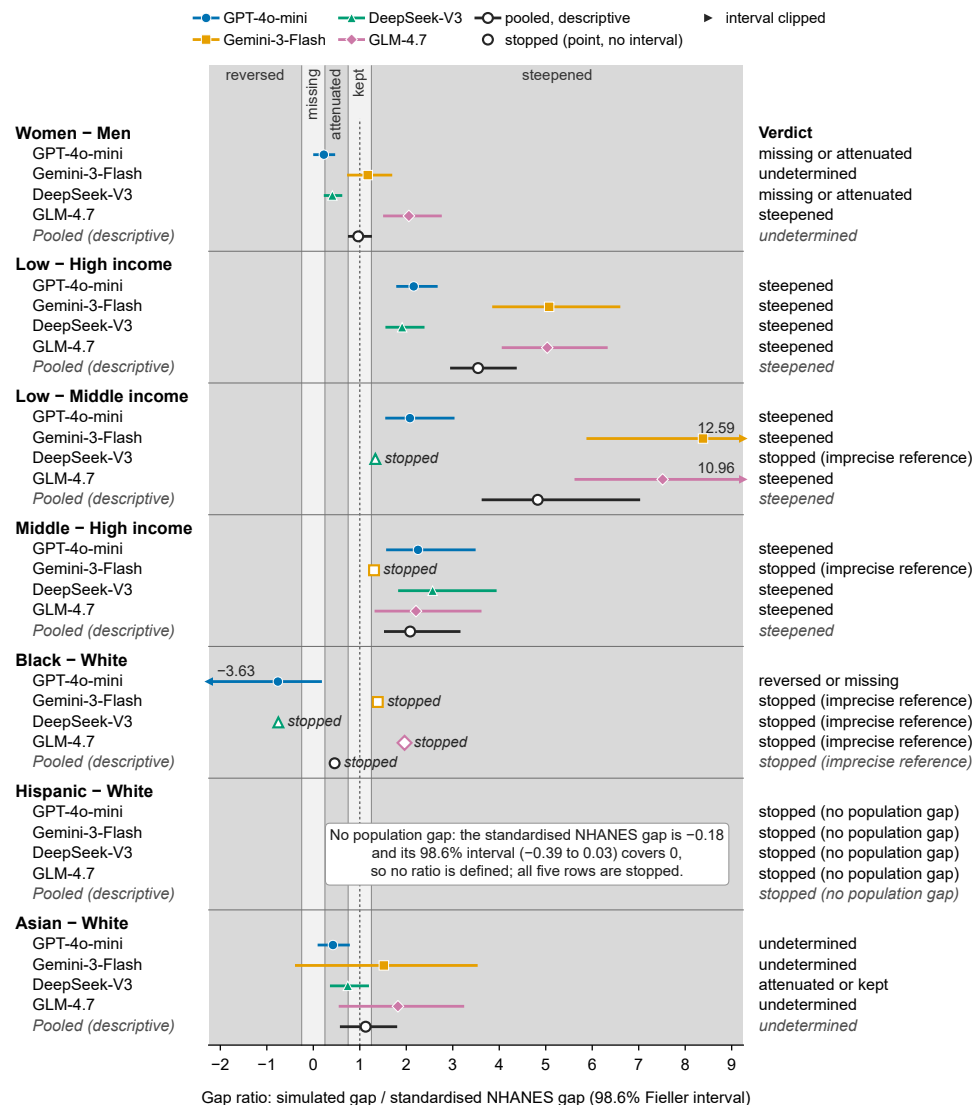
Figure 2 gives the per-model verdicts against the standardised NHANES gaps. The gap between low and high income is steepened in every model, at 1.91 (DeepSeek-V3) to 5.07 (Gemini-3-Flash) times the population gap. Across the three income contrasts, 10 of the 12 per-model readings are steepened. The other two, Gemini-3-Flash’s middle-to-high gap (1.31) and DeepSeek-V3’s low-to-middle gap (1.33), do not exclude kept, and the population intervals for those two gaps are too wide to certify kept (k of .33 and .31), so they record a reference too imprecise. On the marginal reference the income gaps are smaller relative to the population’s, and Low minus High is still steepened for Gemini-3-Flash and GLM-4.7 and kept or steepened for the other two (Supplement S4).

The sex gap divides the models. GPT-4o-mini and DeepSeek-V3 keep a quarter to two fifths of the population’s sex gap (ratios 0.23 and 0.41), read missing or attenuated. GLM-4.7 doubles it (2.06), and Gemini-3-Flash is undetermined. The pooled ratio of 0.97 averages these opposite errors.

No racial contrast receives a single-region verdict in any model. Among adults of equal income, sex and marital status, the NHANES Black–White gap is -0.28 points, while the crude gap is $+0.33$. The standardised Hispanic–White gap is -0.18 , and its interval covers zero, so that contrast records no population gap for every model. The standardised Black–White gap is too imprecise to certify kept ($k = .76$). GPT-4o-mini still reads reversed or missing, and the other three models record a reference too imprecise. For Asian–White, DeepSeek-V3 reads attenuated or kept and the other three are undetermined. The earlier report of these data read the Black–White and Hispanic–White gaps as missing (Keough, 2026). That reading compared conditional simulated gaps with crude survey gaps, and it does not survive the like-for-like reference.

Two checks bear on these verdicts. With the draw-only standard error, 8 of the 35

Level 2: Simulated Subgroup Gaps Against the Standardised NHANES Gaps, per Model



Note. Points are ratios of the simulated PHQ-8 gap to the standardised NHANES 2005–2018 gap, and lines are 98.6% Fieller intervals. Shaded bands are the five regions of the rule, and the dotted line marks a ratio of 1. The pooled Women–Men interval (0.746 to 1.262) crosses both edges of the kept band, so it is undetermined. Pooled rows (hollow circles, grey italic verdicts) are descriptive. Rows stopped as reference too imprecise show a point without an interval, rows with no population gap are not drawn, and arrows mark clipped intervals. Each model has 12 persona pairs per racial contrast, 24 for sex and 16 per income contrast.

verdicts become sharper and none moves outside its primary verdict. Under four other mappings of relationship status in the reference, 25 of the 112 per-model verdicts change, 23 of them on race or sex, and the low-to-high income gap stays steepened in every model (Supplement S3). In the prompt control, the pooled low-to-high and low-to-middle income gaps stay steepened under all three system prompts, so the steepening does not come from the role or the population-reference instruction. With four persona pairs per model in that run, the per-model readings range from undetermined to steepened, and only Gemini-3-Flash’s low-to-high gap is steepened under every prompt. The Controls section shows that at NHANES precision this rung can detect distortion and cannot award a faithful model the verdict KEPT.

In the worked-case stratum (Black women who are single), GPT-4o-mini puts the low-income persona 3.4 points above the high-income one. In NHANES the same stratum differs by 1.6 points.

Level 3: Population Calibration

Every model’s simulated population sits above NHANES (Table 4). After post-stratification the overall residual is +2.1 points for GLM-4.7, +2.6 for Gemini-3-Flash, +4.6 for DeepSeek-V3 and +4.7 for GPT-4o-mini. All 40 model-by-group residuals are positive, with 90% intervals above zero. No model passes all ten groups at 0.2 SD, 1 point or 2 points. Of the 40 rows, none passes at 0.2 SD, one passes at 1 point, and six pass at 2 points. In every model the largest residual is in the low-income band.

Against NHANES 2021–2023, with cohabiting adults counted as married in both cycles, every residual falls by 0.9 to 2.1 points and no model passes all ten groups at 2 points. The overall rows of Gemini-3-Flash and GLM-4.7 pass at 2 points against that cycle, so a use that needs only the national level would pass these two models against 2021–2023 and fail them against 2005–2018. Across all eight declared reference specifications no model passes all ten groups at 2 points, and the most any model reaches is 8 of 10, for GLM-4.7 against 2021–2023. The overall row passes at 2 points for GLM-4.7

Table 4*Level 3: Post-Stratified Residual of the Simulated Population Against NHANES*

Model	Overall residual, points			Groups passing 0.2 SD / 1 / 2 pt	PHQ-8 $\geq$ 10 (%)	
	2005–2018 [90% CI]	2021–2023	Range over 10 groups		Simulated	NHANES
DeepSeek-V3	+4.58 [4.49, 4.67]	+3.47	+4.06 to +5.13	0 / 0 / 0	7.35	7.01
Gemini-3-Flash	+2.59 [2.49, 2.69]	+1.36	+1.33 to +7.37	0 / 0 / 2	16.42	7.01
GPT-4o-mini	+4.68 [4.57, 4.79]	+3.57	+4.23 to +5.32	0 / 0 / 0	16.92	7.01
GLM-4.7	+2.12 [2.00, 2.24]	+0.91	+0.66 to +6.40	0 / 1 / 4	13.23	7.01
Pooled (descriptive)	+3.49 [3.42, 3.57]	+2.33	+2.59 to +6.05	0 / 0 / 0	13.48	7.01

Note. Residual = simulated minus NHANES mean PHQ-8, with each persona mean weighted by the NHANES share of its demographic cell (post-stratification), both framings averaged, over the 48 personas with a reference cell. The 10 groups are all adults, four races, two sexes and three income bands. Groups passing = rows whose 90% interval lies inside ± 0.79 points (0.2 SD), ± 1 point or ± 2 points; a model passes for a use only when all 10 pass. 2021–2023 = the residual against NHANES 2021–2023, analysed with its own design and with cohabiting adults counted as married (the 2005–2018 column leaves them out). The simulated share scoring 10 or more is post-stratified in the same way. Every group row is in Supplement S4.

under four specifications, the two persona-string income crosswalks and the 2017–2020 and 2021–2023 periods, and for Gemini-3-Flash under the 2021–2023 period alone. The model-level failure therefore holds under every specification, while an overall-row pass depends on the reference chosen.

The prevalence row does not follow the mean. Pooled over models, 13.5% of simulated respondents score 10 or more against 7.0% in NHANES. In the low-income band the simulated share is 55.5% against 13.6%. In the middle and high income bands the simulated share is below the population’s (4.4% against 7.3%, and 1.5% against 3.7%), although the means there are too high. In 8 of the 50 primary rows (ten groups for each model and for the pooled population) the mean is significantly too high and the prevalence

significantly too low. DeepSeek-V3’s overall prevalence passes the ratio band (1.05) only because a low-income excess (32.7% against 13.6%) offsets shortfalls in the middle and high bands. The worked-case persona’s mean of 10.0 over both framings sits 5.3 points above her own NHANES cell (4.7). Against adults on Medicaid with household income under \$35,000, the label her prompt carried, the mean of all such adults is 5.5.

Level 4: Structural Fidelity

No model passes level 4 (Table 5). DeepSeek-V3 and GPT-4o-mini have no general factor in either framing. Their ratios of the first to the second eigenvalue are 1.09 to 1.73, against 7.47 in NHANES, and each has negative loadings. The rule fails in every one of 1,000 persona bootstrap replicates for both models. Their PHQ-8 total is therefore not a one-dimensional score, and their later rules are description.

GLM-4.7 and Gemini-3-Flash pass R1, and the pattern of their loadings matches NHANES (congruence .990 to .998). Their loadings are larger than people’s: the mean loading is .83 to .85 for Gemini-3-Flash and .79 to .84 for GLM-4.7, against .76 in NHANES. Gemini-3-Flash fails R2 in both framings (loading RMSD .14 to .15) and GLM-4.7 fails it in the narrative framing (.10, upper limit .11), so the items of these two simulated populations hang together more tightly than NHANES respondents’ answers do. Both models fail R4 on income in both framings. In NHANES, income is invariant at the .08 margin at both steps. In the two models the metric step’s $RMSEA_D$ is .16 to .20, with every lower 90% limit above .13. At the .08 margin, seven of the eight sex steps in these two models are indeterminate, because 24 to 30 personas per group cannot resolve them, and seven of the eight race steps hold, as they do in NHANES. The verdicts are the same at the .05 margin.

The R3 diagnostic shows where the factor in these two models comes from. In NHANES, 3.5% of item variance lies between demographic cells, and the covariance within cells carries the general factor on its own ($\lambda_1/\lambda_2 = 4.33$). In Gemini-3-Flash and GLM-4.7, 48% to 64% of item variance lies between personas, and among draws of one persona there

Table 5*Level 4: Structure of the Simulated Population Against NHANES*

Model	Framing	R1: general factor			R2: loadings		R4: invariance		R3 (diagnostic)	
		λ_1/λ_2	Min. loading		ϕ_T [lower]	RMSD [upper]	at .08	Level 4	Between	Within λ_1/λ_2
DeepSeek-V3	Clinical	1.73	-.72	Fail	.349 [.285]	.755 [.786] (Fail)	–	Fail	.17	1.45
	Narrative	1.43	-.68	Fail	.163 [.074]	.823 [.860] (Fail)	–	Fail	.14	1.34
Gemini-3-Flash	Clinical	8.69	.71	Pass	.990 [.986]	.138 [.150] Fail	Fail	Fail	.63	1.29
	Narrative	9.33	.69	Pass	.990 [.987]	.150 [.159] Fail	Fail	Fail	.64	1.45
GPT-4o-mini	Clinical	1.09	-.24	Fail	.364 [.178]	.718 [.778] (Fail)	–	Fail	.10	1.23
	Narrative	1.33	-.57	Fail	.043 [-.017]	.825 [.846] (Fail)	–	Fail	.06	1.26
GLM-4.7	Clinical	5.73	.66	Pass	.996 [.994]	.081 [.090] Pass	Fail	Fail	.48	1.59
	Narrative	8.12	.75	Pass	.998 [.997]	.100 [.112] Fail	Fail	Fail	.59	1.34
NHANES		7.47	.67						.035	4.33

Note. One-factor model on polychoric correlations of all draws of one model and framing. R1 passes when every loading is at least .30 and $\lambda_1/\lambda_2 \geq 3$. ϕ_T = Tucker’s congruence with the NHANES loadings, with the lower limit of its 90% interval; RMSD = root mean square difference from the NHANES loadings, with the upper limit of its 90% interval. R2 passes when the congruence lower limit is at least .95 and the RMSD upper limit is at most .10. R4 = every metric and scalar invariance step across sex, race and income matching the NHANES verdict at an $RMSEA_D$ margin of .08; it fails on any determinate step that differs from NHANES, is unresolved when no step differs but some are indeterminate, and is not read for a population without a general factor (–). Parenthesised R2 verdicts are descriptive, because R1 failed. Between = share of item variance between personas (NHANES: between demographic cells). Within λ_1/λ_2 = the R1 ratio fitted to the within-persona (NHANES: within-cell) covariance. Level 4 passes on R1, R2 and R4.

is no general factor (λ_1/λ_2 of 1.29 to 1.59). The simulated correlation matrix is built from shifts in persona means, and draws of one persona share almost no common factor, unlike NHANES respondents within one demographic cell. The same uniformity within personas appears at level 1, and it rules out the same single-draw uses.

The earlier report found that these data failed structural fidelity on income through a pooled congruence of .70 (Keough, 2026). That value comes from pooling two models that have a general factor with two that do not. Within each model that has a factor, the income congruence is .961 to .990, and the failure rests on the invariance test instead.

Controls: Real People and Planted Failures

Every model fails every rung above the gate, so the rungs have to be shown to pass real people and to detect failures of a known size.

Design. Each replicate splits the NHANES 2005–2018 adults at random into two halves, stratified by the 48 persona cells. One half supplies respondents and the other half is the reference. A pseudo-model has the corpus design: 120 personas, two framings and 30 draws. Each draw is a respondent from the donor half, drawn within the persona’s cell with probability proportional to the examination weight, and it keeps that respondent’s eight item scores. Transgender and multiracial personas draw from the closest available group and, as in the main analysis, enter levels 1 and 4 only. Four independent pseudo-models form a panel, and there are 200 replicates. Planted failures are applied to the same panels at a grid of doses: shuffling a draw’s item values (incoherence), replacing a draw with the most typical vector at its total (compression), scaling a group gap by a factor g , shifting every persona mean by d points, and rebuilding draws item by item from other draws of the same persona (removing within-persona covariance). Each rung is read with its own rule. Run on the real corpus with the full reference, the control code reproduces every published rung number, 303 of 303 checks.

What real people pass. The level-1 rule is read per framing, and under it real respondents pass both framings in 92.9% of 70 replicates, with the remaining replicates

unresolved and none failed. The pseudo-model passes level 3 on all ten groups in 100% of replicates at 2 points and 98.1% at 1 point. It holds R1 and the R3 diagnostic in both framings in all 200 replicates, and R2 in all 40 replicates of the R2 calibration (loading RMSD upper limit .044 on average). Real respondents are unresolved on R4 at the .08 margin in 24 of 25 replicates and fail it in 1, since 24 to 40 personas per group leave some invariance steps unresolved. Making three items independent within the low-income group produces an R4 failure in 25 of 25 replicates, located at the income metric step. At level 2, a determinate wrong verdict occurs in at most 1.0% of readings at any dose and scope. The level-2 intervals cover the true ratio in .986 to 1.000 of replicates per model, against .986 nominal.

Planted failures are detected. Table 6 gives, for each rung, the dose at which a planted failure is detected in at least 95% of replicates, beside the distortion the real models show. At levels 3 and 4 and for compression at level 1, the real distortion lies beyond the detected dose. At level 2 the control's reference is a half sample, with twice the variance of the full survey, and per-model detection needs a threefold gap. Two models' real income ratios (1.9 and 2.2) lie below that dose, and against the full reference the rung still names both STEEPENED.

What level 2 can and cannot certify. At NHANES precision, level 2 can detect distortion and cannot certify faithfulness. In a simulation at the corpus's own precision, a faithful model is named KEPT alone in at most 37% of readings for the low-to-high income gap and 20% for the sex gap, and in under 1% for every other contrast. These are the most favourable settings, the draw-only standard error at the most precise model, and under the primary pairs standard error neither rate exceeds 2%. For the sex and low-to-high income gaps, a faithful model usually receives a two-region verdict that contains KEPT; for the other five contrasts the population gap is too imprecise to certify KEPT, and the rung records that stop. A twofold steepening of an income or sex gap is named correctly in 87% to 100% of draws under the draw-only standard error. At the audit's own precision, the

Table 6

Detection of Planted Failures, Against the Distortions the Models Show

Rung	Planted failure	Detected in $\geq 95\%$ from	Real models
Level 1	Share of draws with item values shuffled	40% of draws	Not seen
Level 1	Share of draws replaced by the most typical vector at the same total	80% of draws	Misfit ratio 0.00 to 0.18 (both framings); a 100% dose reaches 0.33
Level 2	Income or sex gap scaled by g	$g = 3$ per model (97.4% to 100%)	Low minus High, standardised: 1.9 to 5.1 per model
Level 3	Every persona mean shifted by d points	$d = 2$ at the 2-point tolerance (100%)	+2.1 to +4.7 points overall
Level 4	Share of draws decorrelated within persona	50% fails R2 in 100%; 75% fails R1 in 98.5%	Within-persona λ_1/λ_2 1.23 to 1.59, reached by the planted dose between 75% (1.71) and 100% (1.03)

Note. 200 replicates of a four-model panel of NHANES pseudo-models. Doses are nested, so a draw altered at one dose is altered at every larger dose. Per-dose detection rates for every rung and tolerance are in Supplement S5.

Black–White and Hispanic–White contrasts of a faithful model are stopped in 99.9% of per-model readings. The same design returns these stops for a faithful model, so the stops on race in the Results come from the reference and not from the models.

Intermediate simulators. Table 7 reads seven simulator types, all built from NHANES respondents through the corpus design, at every rung over 70 replicates. Real respondents pass the gate and level 3 in every reading, level 1 in 92.9% of replicates, and R1 and R2 in every replicate, and level 2 fails them in no reading. Each planted failure trips the rung it targets. Independent items fail level 1, R1 and R2 in every replicate, shifted samples fail level 3 in every reading, compressed samples fail level 1 in every replicate, non-invariant samples fail R4 in all 25 replicates, and a doubled income gap fails level 2 in 92.5% of per-model readings. The doubled gap also fails level 3 in 14.6% of readings, because raising the low-income personas raises that group’s level. Two rungs also read plants they do not target. Shifted samples pass the gate in 28.6% of readings at 2.1 points and in none at 4.7 points, because more depressed respondents vary more between draws, so a mean of 30 draws is less precise. R2 passes compressed samples in 1.4% of replicates and non-invariant samples in none, because both change the size of the loadings. Level 4 passes no type, because R4 at margin .08 leaves real respondents unresolved in 24 of 25 replicates, and this design has too few personas per group to resolve invariance.

Larger persona grids. A second simulation repeats the panel over 60 replicates at four persona grids: the corpus’s 48 cells, 144 cells with age or with education added, and 432 cells with both. Real respondents pass the gate’s minimum rule in every reading at every grid. Under the pairs standard error, level 2 fails a faithful simulator in at most 3.3% of per-model readings and fails a doubled income gap in 97.5% to 100%. The draw-only standard error ignores how much real people in one cell differ, and on the finest grid it fails a faithful simulator in 15.4% of readings, which is why the pairs standard error is the primary one. No grid lets level 2 pass a faithful simulator. At 30 draws per persona, a faithful simulator’s low-to-high income gap is never read KEPT under the pairs standard

Table 7

Verdicts for Seven Simulator Types Built From Survey Respondents

Simulator	Gate	L1	L2 fail	L3	R1	R2	R4 fail	R4 unres.	L4
Real respondents	100.0	92.9	0.0	100.0	100.0	100.0	4.0 (25)	96.0	0.0
Independent items	100.0	0.0	0.0	100.0	0.0	0.0	0.0 (5)	0.0	0.0
Shifted, +2.1	28.6	72.9	0.0	0.0	100.0	100.0	0.0 (5)	100.0	0.0
Shifted, +4.7	0.0	15.7	0.7	0.0	100.0	100.0	0.0 (5)	100.0	0.0
Income gap doubled	100.0	84.3	92.5	85.4	100.0	100.0	0.0 (5)	100.0	0.0
Compressed	100.0	0.0	0.0	100.0	98.6	1.4	0.0 (5)	100.0	0.0
Non-invariant (income)	100.0	2.9	0.0	100.0	100.0	0.0	100.0 (25)	0.0	0.0

Note. Percentage of readings that pass, except the columns marked fail or unresolved. Gate, L2 and L3 are read per pseudo-model (280 readings per type); L1, R1 and R2 on one pseudo-model per replicate (70 replicates). R4 is read at margin .08 on the number of replicates in parentheses; it gives no verdict without a general factor. L2 passes in no reading of any type; its remaining readings are unresolved. L4 is the level-4 pass (R1, R2 and R4 in both framings). Shifted types raise every persona by the stated points; the doubled gap raises low-income personas only; non-invariant makes three items independent within the low-income group. Construction, seeds and per-step results: `paper_brm/analysis_brm/intermediate_panel` in the release.

error, whose usual reading is undetermined, and is read KEPT in at most 11.7% of readings under the draw-only one. With no simulation error, which is the limit of adding draws, the same gap is read KEPT in 95.0% and 98.3% of replicates on the 48-cell and the 144-cell age grids. The Black–White and Hispanic–White gaps are never read KEPT on any grid, even with no simulation error, because NHANES measures them too loosely: their median k is .57 to 2.24 against the limit of .25. Certifying a kept gap therefore needs more draws per persona and a population gap the survey measures precisely, and a finer grid does not supply the second: adding education to the grid raises the income contrast’s k from .16 to .24.

Two calibration notes. R2 reads the size of the loadings as well as their pattern. With half the draws decorrelated, the loadings fall from .69 to .52 at their minimum while Tucker’s congruence stays at .999, and the loading RMSD of .17 fails R2 in all 40 replicates. Congruence alone would have passed every one of them, and R1 still holds at that dose. The level-1 tolerance follows from the design. Under the persona design the positive control’s misfit ratio averages 1.15, because the design gives equal weight to cells and over-represents subgroups with more misfit. A tolerance of 1.25 would therefore fail real people drawn through the same design, and the misfit tail passes at $\tau = 1.5$ in 99.4% of replicates.

External Demonstration

We applied the ladder, with the same rules, multiplicity control and stops, to three silicon-sample datasets released by other authors. Two of them have no repeated draws of one persona with a fixed prompt, so the gate and level 1 cannot run on them, and the ladder records those stops. The third has repeated draws, and it was scored after the rules were frozen.

OpinionQA. Meister et al. (2025) released model opinion distributions steered toward demographic groups, alongside the Pew American Trends Panel distributions for the same questions (Santurkar et al., 2023). Both sides are crude group differences, so the reference is marginal. Each model, steering method and attribute is one family. The 154 group contrasts that the earlier version of this protocol had read as 51 kept and 85 missing become, under the ladder’s rule, 87 undetermined, 28 attenuated or kept, 17 missing or attenuated, 10 kept or steepened, 5 reversed or missing, 5 with a reference too imprecise to certify kept, 1 kept and 1 steepened. Most of the earlier “missing” readings become undetermined (48) or two-region verdicts (37), because their intervals span more than one region.

Argyle et al. Study 3. Argyle et al. (2023) released GPT-3 answers conditioned on the backstories of real American National Election Studies respondents, so the simulated

and human gaps are computed on the same people. This removes the question of marginal against standardised reference and makes it the cleanest level-2 case available. In the main run, 23 of 61 contrasts record no population gap and 26 record a reference too imprecise to certify kept. The two Republican–Democrat contrasts on ideology and on the Trump two-party vote share are kept (both 0.92). The Black–White gap in Trump vote share is attenuated (0.53), as is the Republican–Democrat gap in Trump’s share among all voters (0.60). Black–White and Republican–Democrat differences in patriotism are missing. The two kept verdicts show that the rule awards kept when the reference is as precise as it is for these partisan contrasts. Level 3 for both datasets is in Supplement S6, as an illustration, because it depends on use tolerances that the original authors did not state.

Bisbee et al. Bisbee et al. (2024) gave ChatGPT (gpt-3.5-turbo, June 2023, temperature 1) the profiles of 7,530 respondents to the 2016 and 2020 American National Election Studies and asked for a feeling thermometer, from 0 to 100, toward each of 11 groups. Each respondent was asked about 30 times under each of three prompts: the full profile, its demographic part only, and its political part only. Every persona has a human twin, as in Argyle et al., so the reference is the same respondents’ own thermometers. Our cleaning of the released file reproduces the authors’ analysis data, including the 2,235,705 observations and the coefficients of the regression in their Supplementary Section 8 to six decimals. The gate passes the minimum and recommended rules for every thermometer under every prompt, with 1 to 3 draws needed for the minimum rule and 3 to 11 for the recommended rule. Level 2 fails under every prompt. Under the full prompt, 23 of the 77 contrasts record no population gap and 29 a reference too imprecise to certify kept. Three contrasts are kept: the Republican–Democrat gaps in feeling toward the Democratic Party (ratio 1.12) and the Republican Party (1.05), and the Black–White gap in feeling toward Black Americans (0.96). The Republican–Democrat gap is steepened toward five other groups (liberals, Muslims, gays and lesbians, Black Americans and White Americans), with ratios of 1.36 to 2.35. The female–male gaps in feeling toward White Americans and

Christians are reversed or missing, and the age gaps toward the same two groups are attenuated (0.44 and 0.39). Level 3 passes at 0.5 reference SD for 7 of the 11 thermometers under the full prompt and under the political prompt, for 1 under the demographic prompt, and for none at 0.2 SD. With 7,530 respondents the reference is precise enough to certify a kept gap where the population gap is large, as it is for the party gaps on the party thermometers (54 and 57 points). The release also holds 9,238 rows that the authors' code repaired but did not carry into its analysis. Some of them hold misparsed thermometers above 100, so the analysis uses the authors' data; adding the rows changes one of the 231 level-2 verdicts.

Discussion

What the Results Mean for a User

Under this prompt family, the four models should not be used to simulate PHQ-8 respondents. The distortion runs the same way in every model, with population means above the survey's after post-stratification and the low-to-high income gap steepened. Each single draw reads as an answer a person could give, and the distortion appears only across repeated draws and against the survey. Only the rungs above the gate detect the distortion, so a researcher has to run them before using such a sample. The failure belongs to base deployments of these four models, unaltered and under one prompt family, and the paper makes no claim about language models in general. The corpus therefore serves as a baseline, and interventions that ground a model in real population data can be tested against it with the same ladder.

Above the gate, the corpus supports no population use. The gate pass supports reading a persona's mean over enough draws as that model's stable answer for the persona, and that stable answer is what a later intervention would be compared with. The ladder can be run on any clinical silicon sample for which a validated human dataset on the same instrument exists, whether a national survey such as NHANES or a private validated dataset. Without a human reference no rung above the gate can return a verdict, so the

reference has to be chosen before the simulated sample is generated.

Running the Ladder on a New Simulated Sample

The results set four requirements for a researcher who runs the ladder. First, the design needs repeated draws of each persona under a fixed prompt, because the gate and level 1 cannot run without them, and the gate decides how many draws a persona mean needs. Second, the population reference should be computed within the persona design's strata, because a crude survey gap and a conditional simulated gap can differ in sign. Third, a design that tests invariance at level 4 needs more personas per group, since more draws per persona do not narrow the persona-level intervals. Fourth, the tolerance at level 3 is part of the claim and should be named for the use before the residuals are read. The code for every rung is in the release, one script per analysis, and each script writes the receipt files that Supplement S2 maps to the text.

Limitations

Each demographic cell has one persona, so a persona's own quirks cannot be separated from its cell. Transgender and multiracial personas have no NHANES comparison group, so they are not tested at levels 2 and 3. With 24 to 40 personas per group the model failures at levels 2 and 4 are determinate, but a pass at level 4 needs more personas per group, and a pass at level 2 is also limited by the survey's own precision. No clinician reviewed individual answer vectors, and that review is the next test of level 1. The personas carry no age, so the simulated population's age mix is unknown and is not matched to the reference. The PHQ-8 items were presented in one fixed order. Each attribute level was given to the model in one wording, so the verdicts apply to these wordings.

Conclusion

This paper proposes the validity ladder as a protocol that other researchers can apply to any simulated population and extend to new instruments. In the worked example, four models generated 28,800 PHQ-8 assessments, and every model failed every level above

the gate. Pseudo-models built from survey respondents passed the gate, levels 1 and 3, and R1 and R2, and every planted failure tripped the rung it targets. At this design’s size, levels 2 and 4 rarely failed survey respondents (in at most 3.3% and 4% of readings) and certified none: a kept gap needs more draws per persona and a precisely measured population gap, and invariance needs more personas per group. A researcher who uses language models to represent a population should therefore run each rung before reporting a claim that the rung supports. The code and a receipt for every reported number are public, so the ladder can be run on a new corpus without rebuilding it.

Declarations

Funding. No funding was received for this work.

Competing interests. The author has no competing interests to declare.

Ethics approval. Not applicable. The study analyses public, de-identified NHANES microdata and text generated by language models. No human participants were recruited.

Consent to participate and for publication. Not applicable.

Availability of data, materials and code. The corpus, the persona registries, the original generation code for both framings, the analysis scripts and every receipt file are at <https://github.com/pskeough/>

A-Validity-Ladder-for-LLM-Simulated-Clinical-Populations. NHANES data are public from the National Center for Health Statistics, and the release includes the code that extracts them.

Author contributions. Patrick Keough is the sole author and takes responsibility for all content.

Use of generative AI. Anthropic’s Claude models, accessed through Claude Code, were used to implement the analysis and verification scripts to the author’s specification, for LaTeX formatting and typesetting, and to draft prose from the author’s analysis reports and dictated text. The author designed the study, chose the estimands and

the statistical specification, interpreted every result, and revised all text. All generated code was reviewed by the author, and every numeric claim in this paper is checked against its receipt by the verification scripts released with the data. The language models studied in this paper are the objects of the research and are described in the Worked Example: Data section.

Open Practices Statement

The data, materials and analysis code for this study are available at <https://github.com/pskeough/A-Validity-Ladder-for-LLM-Simulated-Clinical-Populations>. The analysis reported here is release commit 799ac17. Supplement S2 maps every number in the body of the article to the file in that repository that prints it and the script that writes the file. The ladder's rules were frozen in LADDER_SPEC.md at commit 91a1b10 (2 October 2026), before any further dataset was analysed; the corpus analyses were not registered in advance. The decoding control's design file, dated before its first call, is in the repository.

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